

NAHA AP, Japan

WMO: 479300

Lat: 26.196N

Lon: 127.646E

Elev: 12

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	54.0	55.6	37.4	32.8	56.9	39.2	35.2	57.8	32.1	58.5	29.7	59.1	17.5	350	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	90.0	79.8	89.6	79.8	88.2	79.8	82.0	86.5	81.4	86.1	80.8	85.8	12.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.7	159.7	86.1	80.3	157.1	85.7	79.1	151.2	84.9	45.8	87.5	45.2	87.1	44.6	87.2	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.8	26.0	23.1	51.5	92.0	2.4	1.7	49.7	93.2	48.3	94.1	46.9	95.1	45.1	96.3		
			44.3	83.0	1.7	0.9	43.1	83.7	42.1	84.2	41.2	84.7	40.0	85.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.7	63.8	64.4	66.9	71.4	76.1	81.2	84.9	84.8	83.0	79.0	73.5	67.1
	DBStd	8.37	4.19	4.70	4.51	3.92	3.26	3.11	1.84	1.97	2.31	3.05	3.64	4.23
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	201	74	63	35	3	0	0	0	0	0	0	0	26
	CDD50	9028	428	404	523	641	810	937	1082	1078	991	898	705	531
	CDD65	3755	37	47	93	194	345	487	617	613	541	433	256	92
	CDH74	34520	14	31	97	523	2055	5083	7929	7825	6273	3580	1005	105
	CDH80	11613	0	0	0	19	230	1493	3504	3409	2203	688	66	1
(14) Wind	WSAvg	11.5	12.6	12.4	12.0	10.9	10.0	11.2	11.1	10.8	11.4	12.0	11.6	12.3
(15) Precipitation	PrecAvg	81.4	4.4	4.4	5.9	6.5	9.6	10.3	6.9	9.9	8.4	6.5	4.7	4.4
	PrecMax	130.9	10.8	13.2	16.5	20.7	25.1	33.9	20.1	26.5	43.1	27.1	14.4	10.2
	PrecMin	38.0	0.9	0.8	1.1	1.3	0.5	0.8	0.2	0.9	0.9	0.3	0.2	0.0
	PrecStd	19.4	2.2	2.8	3.5	4.1	5.7	6.2	5.8	5.8	7.0	5.1	3.0	2.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.5	77.1	78.8	82.2	85.8	89.6	91.7	91.8	91.1	88.1	84.2	78.9
		MCWB	68.1	71.2	72.4	76.2	79.1	80.7	80.0	80.1	79.3	78.4	76.1	71.5
	2%	DB	73.6	75.0	76.7	80.4	84.1	88.0	90.0	90.0	89.5	86.2	82.1	76.9
		MCWB	66.5	68.7	70.2	74.9	77.8	80.7	79.8	79.9	79.2	77.5	74.0	69.1
	5%	DB	71.8	73.3	75.0	78.7	82.5	87.4	89.6	89.6	88.0	84.5	80.4	75.1
		MCWB	64.6	67.3	68.9	73.4	76.4	80.5	79.8	79.9	79.0	76.0	72.6	67.2
	10%	DB	70.1	71.5	73.4	77.0	80.9	86.0	88.2	88.2	87.5	83.9	78.8	73.4
		MCWB	63.2	65.9	67.8	71.9	75.4	80.1	79.9	79.9	78.8	75.5	71.8	66.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.7	72.9	73.4	77.2	80.0	82.6	82.7	82.7	81.8	80.0	77.8	73.1
		MCDB	74.1	75.4	76.6	80.2	83.2	86.4	87.0	87.3	86.7	85.0	82.0	76.8
	2%	WB	68.2	70.6	72.1	75.8	78.9	81.7	81.9	81.9	80.7	79.0	75.8	71.1
		MCDB	72.0	73.5	75.1	78.9	82.1	85.5	86.6	86.7	86.1	83.9	79.8	74.7
	5%	WB	65.9	68.7	70.6	74.5	78.1	80.9	81.3	81.2	80.1	78.1	74.5	69.0
		MCDB	70.5	72.0	73.7	77.6	81.2	84.8	86.3	86.4	85.4	83.1	78.7	73.3
	10%	WB	64.0	66.6	68.9	73.0	76.9	80.3	80.7	80.6	79.6	77.1	72.9	67.1
		MCDB	69.2	70.6	72.3	76.1	80.1	84.1	86.1	86.1	84.9	82.1	77.8	72.3
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	7.7	8.0	7.8	7.3	6.6	7.0	7.1	7.3	7.3	7.3	7.4
		MCDBR	9.2	9.1	8.9	8.4	7.8	6.9	7.7	8.0	8.1	8.1	8.0	8.7
		MCWBR	7.1	6.9	6.8	6.0	4.9	3.4	3.5	3.6	3.9	4.5	5.2	6.2
	5% WB	MCDBR	8.3	8.2	8.1	7.8	7.1	6.6	6.8	7.2	7.3	7.4	7.5	7.7
		MCWBR	7.8	7.8	8.2	6.6	5.4	4.0	3.8	3.9	4.0	4.6	5.9	7.0
(40) Clear-Sky Solar Irradiance	taub		0.462	0.491	0.565	0.573	0.516	0.473	0.474	0.492	0.487	0.486	0.469	0.461
	taud		2.126	2.064	1.892	1.932	2.157	2.340	2.332	2.270	2.253	2.196	2.201	2.134
	Ebn at Noon		245	248	237	239	252	261	261	256	255	247	242	239
	Edh at Noon		44	50	62	61	49	40	41	43	43	43	40	42
(44) All-Sky Solar Radiation	RadAvg		772	976	1204	1405	1569	1737	2063	1889	1657	1306	956	780
	RadStd		92	147	114	143	202	157	168	129	128	114	88	77

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	+0.65	+0.76	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	+0.74	+0.95	N/A	N/A	N/A	N/A

Nomenclature: See separate page